

SODANKYLA VUOTSO, Finland

WMO: 028160

Lat: 68.084N

Lon: 27.185E

Elev: 249

StdP: 98.37

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.4	-37.2	0.1	-33.6	-33.6	0.2	-30.4	6.4	-4.5	5.7	-4.5	1.0	340	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	24.9	15.8	22.3	15.0	20.1	14.2	17.3	22.2	16.1	20.7	15.0	18.9	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.4	18.3	14.3	10.5	17.2	13.3	9.8	16.3	49.5	22.2	45.9	20.9	42.8	19.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.9	4.3	-37.8	26.9	1.9	2.7	-39.1	28.9	-40.3	30.4	-41.4	32.0	-42.8	33.9		
			-37.4	18.7	2.4	1.8	-39.1	20.0	-40.5	21.0	-41.8	22.1	-43.6	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-14.4	-12.9	-8.9	-2.0	4.8	9.9	13.6	11.0	6.6	-1.0	-7.0	-10.4
	DBStd	10.99	7.47	8.21	6.27	4.75	4.52	4.27	3.99	3.70	3.51	4.63	6.23	7.23
	HDD10.0	4208	756	641	586	361	171	53	12	33	112	341	511	632
	HDD18.3	7006	1014	874	844	611	418	255	154	231	353	599	761	891
	CDD10.0	255	0	0	0	0	11	51	122	63	9	0	0	0
	CDD18.3	10	0	0	0	0	0	2	7	2	0	0	0	0
	CDH23.3	137	0	0	0	0	4	21	93	19	0	0	0	0
	CDH26.7	21	0	0	0	0	0	2	15	4	0	0	0	0
Wind	WSAvg	2.1	1.9	2.1	2.3	2.2	2.4	2.5	2.1	2.1	2.1	1.9	1.9	2.0
Precipitation	PrecAvg	520	30	30	28	32	38	64	67	65	47	50	42	34
	PrecMax	675	74	79	60	98	84	109	121	131	95	104	71	63
	PrecMin	392	9	9	5	9	11	9	19	16	8	10	7	5
	PrecStd	86	15	18	14	19	16	26	26	29	24	27	18	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	2.5	4.4	11.3	23.0	26.2	28.8	25.9	17.6	8.7	4.4	2.1
		MCWB	-0.6	1.5	1.5	6.6	13.1	16.7	17.6	17.7	12.9	7.1	4.0	1.5
	2%	DB	-1.5	0.4	2.5	7.5	18.8	22.5	26.0	22.1	15.6	7.6	1.8	0.8
		MCWB	-1.8	-0.2	0.2	3.5	11.1	14.5	16.0	15.7	11.8	6.6	1.6	0.4
	5%	DB	-2.7	-0.6	0.9	5.7	15.3	19.7	23.5	19.7	14.1	6.3	0.9	0.0
		MCWB	-3.0	-1.1	-0.4	2.4	9.3	13.0	15.4	14.7	10.9	5.4	0.7	-0.2
	10%	DB	-4.5	-2.1	-0.2	4.1	12.4	17.3	21.1	17.4	12.5	4.9	0.2	-1.2
		MCWB	-4.8	-2.5	-1.3	1.3	8.0	11.7	14.7	13.5	10.3	4.4	0.0	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	1.7	2.0	6.5	13.6	17.7	19.5	18.5	14.1	8.0	4.2	1.6
		MCDB	-0.2	2.5	4.0	11.0	21.7	24.2	23.4	24.3	16.6	8.1	4.2	2.0
	2%	WB	-1.8	0.0	0.8	3.9	11.9	16.0	17.8	16.5	12.7	6.7	1.6	0.5
		MCDB	-1.6	0.3	2.0	6.9	17.2	21.6	22.6	20.4	14.5	7.4	1.8	0.7
	5%	WB	-2.9	-1.2	-0.3	2.7	9.9	14.2	16.6	15.2	11.6	5.5	0.7	-0.2
		MCDB	-2.7	-0.7	0.7	5.1	14.1	17.5	21.5	18.7	13.4	6.2	0.8	0.0
	10%	WB	-4.8	-2.5	-1.2	1.9	8.2	12.5	15.6	14.1	10.5	4.4	0.1	-1.5
		MCDB	-4.6	-2.1	-0.2	3.7	12.0	16.5	20.0	16.6	12.1	4.9	0.2	-1.2
Mean Daily Temperature Range	5% DB	MDBR	8.8	8.6	11.3	9.6	8.8	8.7	9.2	9.2	7.7	5.4	6.8	8.2
		MCDBR	8.4	6.3	7.9	10.0	13.9	12.6	13.9	12.9	11.2	5.8	4.4	5.2
		MCWBR	8.4	5.9	6.6	6.5	7.4	5.8	5.8	7.0	7.4	5.0	4.5	5.2
	5% WB	MCDBR	8.2	6.1	6.7	9.1	12.8	10.5	10.7	10.8	8.6	5.2	4.4	5.2
		MCWBR	8.3	5.8	5.8	6.1	7.3	5.5	5.4	6.4	6.2	4.6	4.5	5.2
Clear-Sky Solar Irradiance	taub	0.190	0.249	0.287	0.318	0.337	0.339	0.351	0.333	0.305	0.267	N/A	N/A	
	taud	1.916	2.080	2.159	2.209	2.400	2.480	2.465	2.491	2.545	2.301	N/A	N/A	
	Ebn at Noon	98	541	741	826	856	865	839	816	754	535	N/A	0	
	Edh at Noon	9	47	80	103	96	92	90	79	58	38	N/A	0	
All-Sky Solar Radiation	RadAvg	0.04	0.50	2.00	3.85	4.65	4.96	4.53	3.26	1.71	0.56	0.09	0.00	
	RadStd	0.00	0.06	0.13	0.24	0.45	0.42	0.48	0.30	0.17	0.10	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page